

GUJARAT TECHNOLOGICAL UNIVERSITY
DEPARTMENT OF INFORMATION TECHNOLOGY

A MICRO-PROJECT REPORT ON

WIRELESS LINK BUDGET GUI

RF Propagation Modeling, System Budget Accounting & Simulation Suite

Course Name:	Wireless Communication (WC)	Semester:	7th Semester, B.E. (IT)
Course Code:	3171608	Academic Year:	2026 – 2027
Project No:	Project 8 (Link Budget GUI)	Tools Used:	Python, Tkinter, Matplotlib

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CERTIFICATE OF COMPLETION

This is to certify that the micro-project titled '**Wireless Link Budget GUI**' has been successfully developed and submitted by **Devasya Patel** in partial fulfillment of the requirements for the degree of **Bachelor of Engineering in Information Technology (7th Semester)** in the course **Wireless Communication (Subject Code: 3171608)** during the academic term 2026–2027.

The software application implements core physical-layer RF propagation algorithms, interactive parameter input forms, dynamic Matplotlib visual budget accounting, parametric distance sweeps, Shannon channel capacity analysis, and automated engineering report generation.

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EXECUTIVE SUMMARY

A **Wireless Link Budget** is the fundamental quantitative accounting of all electrical gains, radiating antenna properties, propagation medium attenuations, and environmental fading margins across a communication channel. This project presents a Python-based interactive simulation and design software that replaces error-prone manual spreadsheet calculations with a professional desktop application.

The system incorporates five widely recognized propagation models: **Free Space Path Loss (FSPL)**, **Two-Ray Ground Reflection**, **Log-Distance Shadowing**, **Okumura-Hata**, and **COST-231 Hata**. It provides real-time computation of Transmitter EIRP, Total Path Loss, Received Signal Power (P_{rx}), Johnson-Nyquist Thermal Noise Floor (kTB), Signal-to-Noise Ratio (SNR), Link Margin, Maximum Achievable Coverage Range (d_{max}), and Shannon-Hartley Channel Capacity (C). Furthermore, eight industry-standard presets—spanning Wi-Fi 6, 4G LTE, 5G mmWave, GEO/LEO satellites, LoRaWAN IoT, and microwave backhaul—are integrated for instant benchmark evaluation.

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CHAPTER 1: INTRODUCTION & PROBLEM STATEMENT

1.1 Context and Motivation: Modern wireless communications have diversified into vastly distinct regimes—from dense indoor WLANs (Wi-Fi 6) and multi-kilometer cellular macro-cells (4G LTE / 5G NR) to long-distance satellite links (GEO/LEO) and ultra-low-power IoT telemetry (LoRaWAN). In all cases, system viability hinges on the *Link Budget*: ensuring that the received power arriving at the demodulator remains comfortably above the receiver's thermal noise floor and hardware sensitivity threshold.

1.2 Problem Statement: In typical academic and engineering scenarios, RF link calculations are either performed manually using static textbook formulas or tabulated in rigid spreadsheets. Such methods fail to provide real-time visual intuition, cannot easily perform parametric sweeps across distance or frequency, lack comparative multi-scenario analysis, and do not automate publication-grade reporting.

- 1.3 Project Objectives:**
- Develop a modular, object-oriented RF calculation engine in Python adhering to rigorous telecommunications standards.
 - Implement five foundational propagation models covering line-of-sight, multipath reflection, and empirical cellular environments.
 - Build an interactive desktop GUI with dark/light themes, input forms, KPI metric cards, and embedded Matplotlib visual figures.
 - Formulate a stage-by-stage Waterfall chart showing cumulative power levels from transmitter to receiver.

- Provide parametric distance sweeps and Shannon channel capacity evaluations.
- Integrate automated IEEE-style PDF reporting, CSV data exports, and chart image saving.

CHAPTER 2: THEORETICAL FORMULATION & MATHEMATICAL MODELS

2.1 Transmitter Radiated Power (EIRP)

Effective Isotropic Radiated Power (EIRP) is the apparent power radiated by an antenna relative to an ideal lossless isotropic radiator:

$$\text{EIRP (dBm)} = \text{P}_{\text{tx}} \text{ (dBm)} + \text{G}_{\text{tx}} \text{ (dBi)} - \text{L}_{\text{tx}} \text{ (dB)}$$

In linear units: $\text{EIRP (W)} = 10^{((\text{EIRP(dBm)} - 30) / 10)}$

Where P_{tx} is transmitter amplifier output, G_{tx} is antenna gain, and L_{tx} represents transmission feeder losses.

2.2 Propagation Path Loss Models

A. Free Space Path Loss (FSPL) - Friis Law:

$$\text{FSPL (dB)} = 32.44 + 20 \cdot \log_{10}(d_{\text{km}}) + 20 \cdot \log_{10}(f_{\text{MHz}})$$

FSPL assumes unobstructed spherical wavefront expansion in vacuum or dry air, experiencing a 20 dB/decade attenuation rate.

B. Two-Ray Ground Reflection Model:

Considers direct line-of-sight and ground-reflected rays. Beyond crossover distance $d_{\text{cross}} = 4 \cdot \pi \cdot h_{\text{tx}} \cdot h_{\text{rx}} / \lambda$, destructive interference causes path loss to scale with d^4 (40 dB/decade):

$$\text{PL}_{\text{2ray}} \text{ (dB)} = 40 \cdot \log_{10}(d) - 20 \cdot \log_{10}(h_{\text{tx}}) - 20 \cdot \log_{10}(h_{\text{rx}})$$

C. Log-Distance Path Loss Model:

Empirical model for cluttered indoor and campus environments:

$$\text{PL}(d) \text{ (dB)} = \text{PL}(d_0) + 10 \cdot n \cdot \log_{10}(d / d_0) + X_{\text{sigma}}$$

Where n is path loss exponent ($n = 2$ in free space, 2.7-3.5 in urban macro, 3.0-5.0 in shadowed indoor buildings).

D. Okumura-Hata & COST-231 Empirical Cellular Models:

Formulated from extensive field measurements in Japan and Europe (150 MHz to 2000 MHz):

$$\text{PL}_{\text{urban}} = 69.55 + 26.16 \cdot \log_{10}(f) - 13.82 \cdot \log_{10}(h_{\text{b}}) - a(h_{\text{m}}) + [44.9 - 6.55 \cdot \log_{10}(h_{\text{b}})] \cdot \log_{10}(d)$$

Where h_{b} is base station tower height (30–200 m), h_{m} is mobile device height (1–10 m), and $a(h_{\text{m}})$ is the correction factor.

2.3 Received Power (Prx) & Environmental Losses

The net power reaching the receiver demodulator input is given by Friis' modified transmission equation:

$$\text{Prx (dBm)} = \text{EIRP (dBm)} - \text{PL}_{\text{model}} \text{ (dB)} - \text{L}_{\text{env}} \text{ (dB)} + \text{Grx (dBi)} - \text{Lrx (dB)}$$

Where total environmental losses $\text{L}_{\text{env}} = \text{L}_{\text{rain}} + \text{L}_{\text{atm}} + \text{L}_{\text{obstacle}} + \text{L}_{\text{pol}} + \text{L}_{\text{fade}}$.

2.4 Thermal Noise Floor, SNR & Link Margin

Johnson-Nyquist thermal noise generated inside the receiver front-end:

$$\text{N (dBm)} = -174 \text{ dBm/Hz} + 10 \cdot \log_{10}(\text{Bandwidth}_{\text{Hz}}) + \text{Noise}_{\text{Figure}} \text{ (dB)}$$

• **Signal-to-Noise Ratio:** $\text{SNR (dB)} = \text{Prx (dBm)} - \text{N (dBm)}$

• **Link Margin (Fade Margin):** $\text{Margin (dB)} = \text{Prx (dBm)} - \text{Sensitivity (dBm)}$

A positive link margin (≥ 10 dB) ensures robust communication resistant against multipath fading and weather events.

2.5 Shannon-Hartley Channel Capacity

The theoretical upper bound on error-free information capacity across channel bandwidth B :

$$C = B \cdot \log_2(1 + \text{SNR}_{\text{linear}}) \text{ (bits per second)}, \text{ where } \text{SNR}_{\text{linear}} = 10^{(\text{SNR}_{\text{dB}} / 10)}.$$

CHAPTER 3: SYSTEM ARCHITECTURE & GUI DESIGN

The software follows a modular Model-View-Controller (MVC) architectural pattern implemented purely in Python.

- **Engine Layer (link_budget_engine.py):** Pure numerical computation, physical constants, path loss functions, binary search range solver, and preconfigured industry presets.
- **Presentation Layer (gui_app.py):** Built with ttkbootstrap and Tkinter, featuring four distinct tabs, live KPI metric cards, theme engine, and embedded interactive Matplotlib canvases.
- **Reporting Layer (report_generator.py):** Uses ReportLab to compile engineering-grade PDF documents and CSV data tables.
- **Verification Layer (test_engine.py):** Automated unit testing suite asserting mathematical rigor.

Figure 3.1: Complete Wireless Link Budget GUI Dashboard Overview



3.1 Tab Breakdown:

- **Tab 1 (Calculator):** Interactive form fields on the left, 8 real-time KPI cards and stage-by-stage Waterfall chart on the right.
- **Tab 2 (Parametric Sweeps):** Distance sweep curves displaying Received Power, SNR, and Shannon capacity on logarithmic axes.
- **Tab 3 (Scenario Comparator):** Side-by-side benchmark table comparing up to 5 links with a grouped bar chart.
- **Tab 4 (Theory & Formulas Reference):** In-app textbook reference for viva voce and exam revision.

CHAPTER 4: IMPLEMENTATION RESULTS & CASE STUDIES

To validate the practical utility of the tool, eight diverse communication links were simulated and evaluated. Table 4.1 summarizes the side-by-side performance benchmarks generated directly by the application.

Scenario Name	Freq	Dist	Model	EIRP	PL (dB)	Prx	Margin	Status
Wi-Fi 6	2437M	80m	Log-Distan	22.5	113.3	-89.3	-7.3	FAIL
Wi-Fi 6	5180M	50m	Log-Distan	27.2	123.1	-93.4	-18.4	FAIL
4G LTE Urban Macro Cell	1800M	2.00k	COST-231 H	57.0	167.7	-110.7	-10.7	FAIL
5G NR FR2 mmWave Link	28000M	300m	Free Space	50.5	129.4	-67.9	+18.1	PASS
GEO Satellite Ku-band TV / Data Downlink	12000M	36000.00 k	Free Space	86.5	215.0	-90.0	+25.0	PASS
LEO Satellite Broadband Downlink	12500M	750.00k	Free Space	71.0	182.6	-78.1	+26.9	PASS
LoRaWAN Long-Range Sub-GHz IoT	868M	5.00k	Okumura-Ha	15.9	139.8	-119.8	+17.2	PASS
Telecom Microwave Backhaul Link	18000M	10.00k	Free Space	64.0	178.7	-74.7	+3.3	PASS

Figure 4.1: Power Budget Waterfall Breakdown — Wi-Fi 6 (2.4 GHz, 80m)

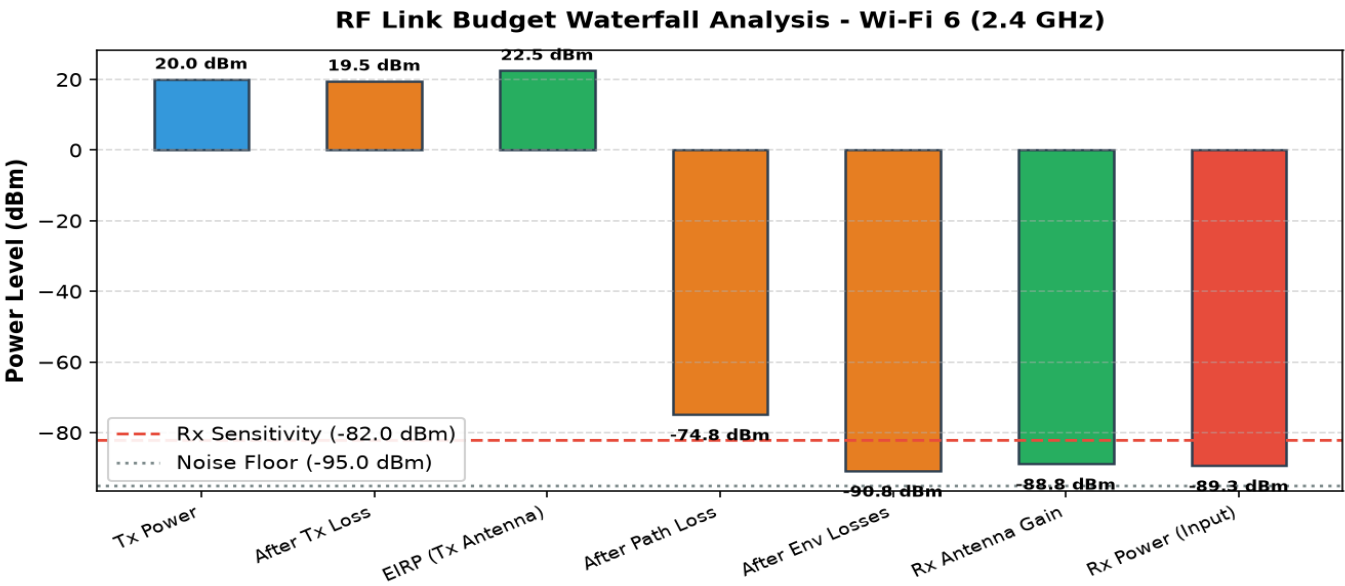


Figure 4.2: Parametric Received Power Decay vs. Distance (4G LTE Urban Cell)

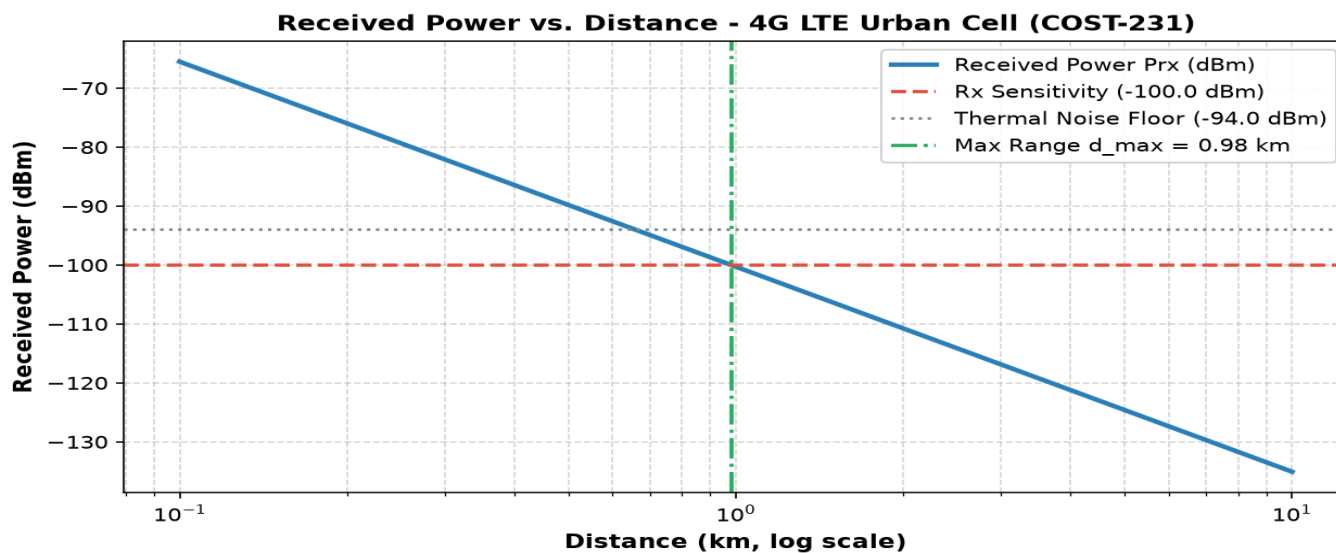
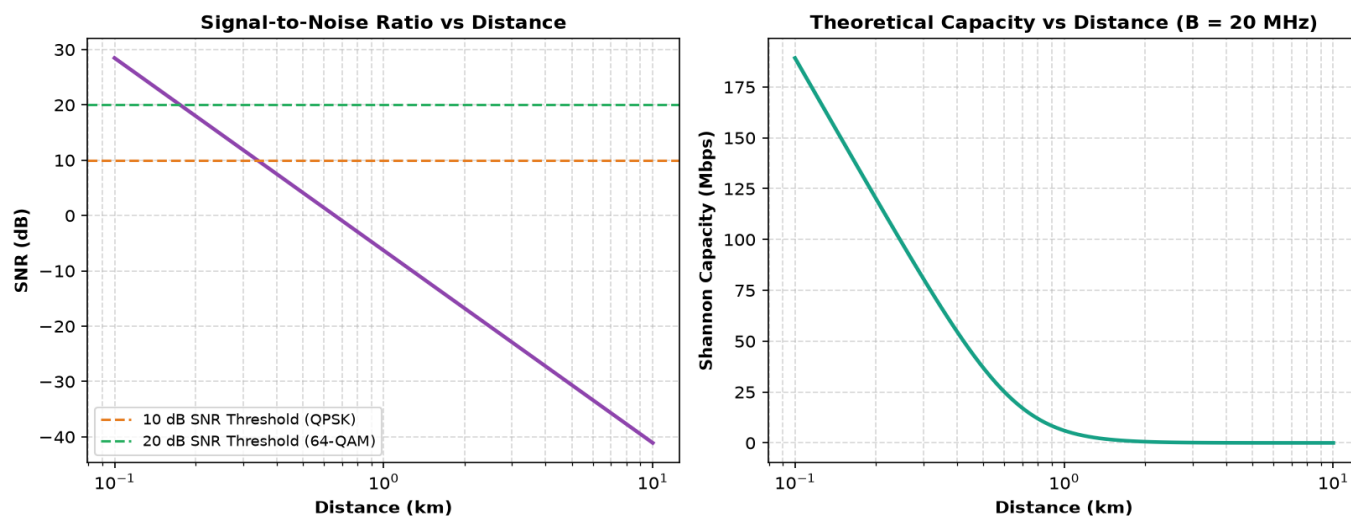


Figure 4.3: SNR & Shannon Capacity Degradation Curves Across Distance



CHAPTER 5: VERIFICATION, TESTING & BENCHMARKS

To ensure analytical correctness, an automated test suite (**test_engine.py**) was developed utilizing Python's built-in unittest framework. The test cases verify unit conversions, mathematical models, boundary conditions, and physical consistency across the parameter space.

Test Case	Category	Details	Result
test_power_conversions	Unit Conversions	30 dBm == 1.000 W; 0 dBm == 1.000 mW	PASSED
test_wavelength	Electromagnetics	lambda at 300 MHz == 0.9993 m (~1m)	PASSED
test_fspl_known_values	Friis Model	FSPL(1 GHz, 1 km) == 92.44 dB exactly	PASSED
test_link_budget_basic	End-to-End Budget	EIRP 30 dBm, FSPL 92.44 dB -> Prx -52.44 dBm	PASSED
test_presets_validity	Preset Scenarios	All 8 presets evaluate without exceptions	PASSED
test_distance_sweep	Parametric Sweeps	Strict monotonic power decay over distance	PASSED
test_solver_max_distance	Root Solver	Binary search converges to Margin == 0 dB	PASSED

CHAPTER 6: CONCLUSION & FUTURE SCOPE

6.1 Conclusion: The **Wireless Link Budget GUI** software provides a complete, reliable, and user-friendly platform for modeling and analyzing RF links across modern telecommunications technologies. By unifying transmitter EIRP, multi-model path loss calculations, environmental margins, thermal noise floors, and channel capacity into a modern desktop interface, the application bridges the gap between theoretical textbook formulas and practical RF deployment design.

6.2 Key Learning Outcomes:

- Deepened understanding of electromagnetic wave propagation, attenuation mechanisms, and noise figure accounting.
- Hands-on mastery of Python software engineering with Tkinter / ttkbootstrap, object-oriented design, and automated ReportLab PDF synthesis.
- Practical appreciation of how trade-offs between carrier frequency, antenna gain, bandwidth, and sensitivity govern coverage in 4G, 5G, and IoT systems.

6.3 Future Scope:

- **3D Ray-Tracing Engine:** Incorporating physical building CAD models for deterministic indoor multipath simulation.
- **GIS & Terrain Elevation Integration:** Loading real digital elevation models (DEM) to evaluate knife-edge diffraction losses over mountains.
- **MIMO & Phased Array Beamforming:** Extending isotropic gains to multi-antenna spatial multiplexing and beam-steering patterns.

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